

PARAINEN FAGERHOLM, Finland

WMO: 029500

Lat: 60.112N

Lon: 21.698E

Elev: 15

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	3.3	9.3	-1.3	5.1	3.6	4.4	6.9	11.4	35.2	33.5	31.8	33.9	7.7	40	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	76.7	68.5	73.5	66.6	70.8	64.6	70.7	74.9	68.0	72.0	65.7	69.4	10.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.1	107.0	73.1	66.5	97.7	70.4	64.2	90.1	67.9	34.6	74.3	32.4	72.1	30.6	69.4	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.6	27.9	24.8	3.7	80.6	10.8	4.5	-4.1	83.8	-10.4	86.4	-16.4	88.9	-24.2	92.1		
			2.6	70.1	11.1	4.4	-5.3	73.3	-11.8	75.8	-18.0	78.3	-26.1	81.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.9	28.3	27.8	31.2	38.7	49.5	57.3	64.5	63.3	56.3	46.4	40.1	34.4	(6)
	DBStd	14.16	7.87	9.33	6.32	4.55	6.53	4.77	5.03	3.90	4.55	5.05	6.25	7.25	(7)
	HDD50	3234	672	623	582	342	91	3	0	0	8	131	297	485	(8)
	HDD65	7419	1137	1043	1047	789	481	234	73	79	261	578	747	950	(9)
	CDD50	1376	0	0	0	2	75	223	449	412	197	18	0	0	(10)
	CDD65	88	0	0	0	0	0	4	57	26	1	0	0	0	(11)
	CDH74	248	0	0	0	0	1	14	193	40	0	0	0	0	(12)
	CDH80	30	0	0	0	0	0	0	27	3	0	0	0	0	(13)
(14) Wind	WSAvg	12.1	13.2	12.2	12.1	10.3	10.0	11.1	10.5	11.2	11.7	13.2	14.7	14.9	(14)
(15) Precipitation	PrecAvg	24.4	2.2	1.6	1.5	1.3	1.1	1.9	2.1	2.6	2.4	2.9	2.8	2.4	(15)
	PrecMax	29.9	3.7	3.3	3.4	2.2	2.8	3.1	4.3	4.8	7.4	5.7	6.1	3.9	(16)
	PrecMin	18.7	0.4	0.1	0.2	0.2	0.2	0.6	0.2	0.3	0.4	1.3	0.4	0.5	(17)
	PrecStd	2.6	0.9	0.9	0.7	0.6	0.7	0.7	1.3	1.2	1.8	1.0	1.5	1.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	41.4	45.6	57.8	71.5	76.0	83.5	79.3	68.4	57.5	51.1	47.1	(19)
		MCWB	39.8	38.0	38.6	48.0	58.8	63.4	70.1	70.3	64.3	53.8	49.1	44.5	(20)
	2%	DB	40.3	39.3	42.5	52.3	67.2	70.9	79.3	73.9	65.4	55.5	49.8	45.0	(21)
		MCWB	38.9	37.4	38.4	45.0	57.4	61.5	69.1	67.8	61.3	53.3	48.5	43.0	(22)
	5%	DB	39.1	37.8	40.4	48.3	63.9	67.9	75.4	71.3	63.8	54.1	48.5	43.6	(23)
		MCWB	37.9	36.6	37.3	42.4	55.9	60.1	68.4	65.4	60.1	51.9	47.2	41.8	(24)
	10%	DB	37.7	36.7	38.5	45.3	60.2	65.2	72.7	69.1	62.3	52.9	47.0	42.3	(25)
		MCWB	36.4	35.7	36.0	40.8	53.5	58.6	66.7	63.8	58.8	50.8	45.3	40.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.5	39.5	41.6	49.2	61.8	67.4	74.8	73.7	65.2	55.4	50.1	45.3	(27)
		MCDB	40.9	40.6	44.0	56.0	67.3	71.9	76.5	75.1	66.6	56.2	50.4	46.3	(28)
	2%	WB	39.3	37.8	39.0	45.6	58.7	63.4	72.0	69.2	62.6	53.9	48.6	43.6	(29)
		MCDB	40.0	38.6	41.0	50.8	64.9	68.7	76.8	72.7	64.5	54.8	49.6	44.5	(30)
	5%	WB	38.1	36.8	37.7	43.2	56.4	61.4	69.7	67.0	60.9	52.8	47.3	42.1	(31)
		MCDB	38.9	37.6	39.5	47.1	62.7	66.1	73.9	70.1	63.0	53.8	48.4	43.1	(32)
	10%	WB	36.6	35.8	36.5	41.5	54.0	59.7	67.2	65.1	59.5	51.3	45.6	40.8	(33)
		MCDB	37.6	36.6	38.2	44.6	59.5	63.9	71.5	67.7	61.6	52.5	46.7	42.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	6.9	8.6	10.1	11.0	9.6	8.9	7.4	6.6	5.5	4.7	5.3	(35)
		MCDBR	5.1	5.9	9.2	15.7	16.7	14.9	13.6	10.3	8.1	5.5	5.0	5.5	(36)
	5% WB	MCWBR	5.0	5.0	6.5	9.1	8.9	7.7	6.3	5.3	5.4	5.0	4.9	5.4	(37)
		MCWBR	5.1	5.7	8.0	13.6	15.8	12.9	11.1	7.9	7.1	5.3	5.0	5.0	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.281	0.315	0.333	0.372	0.368	0.361	0.389	0.380	0.351	0.335	0.307	0.393	(40)
		taud	2.174	2.249	2.309	2.319	2.418	2.478	2.423	2.437	2.502	2.404	2.216	2.637	(41)
	Ebn at Noon	Ebn at Noon	153	214	252	262	272	277	264	257	247	208	139	107	(42)
		Edh at Noon	13	22	28	33	33	31	32	30	24	19	13	10	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	75	260	751	1281	1761	1922	1793	1400	858	365	100	47	(44)
	RadStd	RadStd	11	46	86	126	156	128	157	124	99	54	21	6	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (56 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	-230	N/A	N/A	N/A

Nomenclature: See separate page